



Qualification Pack



Assistant - Pipeline Fitter & Maintenance (Metal Sector)

QP Code: ISC/Q0813

Version: 2.0

NSQF Level: 3

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ISC/Q0813: Assistant - Pipeline Fitter & Maintenance (Metal Sector)

Brief Job Description

The job is to maintain pipe lines across steam line, water line, gas line etc. across the plant. This also involves up-keeping, monitoring and maintenance of pipe lines.

Personal Attributes

This job requires the individual to work independently as well as in teams. He should be physically fit, not having colour blindness, having analytical skills, problem solving attitude, high concentration levels and willingness to work in a factory environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0009: Work effectively with others](#)
3. [ISC/N0820: Understand the assigned pipeline job](#)
4. [ISC/N0821: Prepare for assigned pipeline job operation](#)
5. [ISC/N0822: Carry out the assigned pipe line operation](#)
6. [ISC/N0823: Activities specific to pipeline operation](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Mechanical Maintenance, Non-Ferrous
Occupation	Mechanical Maintenance
Country	India
NSQF Level	3



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Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7233.0100
Minimum Educational Qualification & Experience	10th Class with 6 Months of experience in relevant field OR 8th Class with 3 Years of experience in relevant field OR 5th Class with 7 Years of experience in relevant field OR Certificate-NSQF (Level 2) with 3 Years of experience in relevant field
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQF Approval Date	07/10/2025
Version	2.0
Reference code on NQR	QG-03-IS-045372025-V2-IISSSC
NQR Version	01



Qualification Pack

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about following safety and adopting sustainable practices for optimising use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Emergencies, rescue and first aid procedures
- Health and hygiene
- Housekeeping and waste management
- Material and energy conservation

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** follow safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** use appropriate protective clothing/equipment for specific tasks and work
- PC4.** follow appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC5.** lift heavy objects safely using correct procedures
- PC6.** carry out routine check of the machine for identifying potential hazards
- PC7.** report any identified breaches in health, safety and security policies and procedures to the designated person

Emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC8.** use appropriate type of fire extinguisher
- PC9.** apply appropriate rescue techniques during fire hazard
- PC10.** provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.
- PC11.** follow emergency procedures such as raising alarm, safe evacuation etc.
- PC12.** attend safety training and fire drills to respond promptly during an emergency

Health and hygiene

To be competent, the user/individual on the job must be able to:

- PC13.** follow regular cleaning and disinfection practices at work place using appropriate techniques and materials
- PC14.** follow hand hygiene practices at work place using appropriate techniques and materials
- PC15.** report regarding the contagious illness of self or people in close contact
- PC16.** avoid contact with ill people and self-isolate in a similar situation



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Housekeeping and waste management

To be competent, the user/individual on the job must be able to:

- PC17.** follow the fundamentals of 5S for housekeeping
- PC18.** ensure good housekeeping in order to prevent hazards and accidents
- PC19.** store the material, tools and equipment in the correct location and in good condition
- PC20.** segregate waste into different categories
- PC21.** identify recyclable, non-recyclable and hazardous waste
- PC22.** dispose non-recyclable, recyclable and reusable waste appropriately at identified location

Material and energy conservation

To be competent, the user/individual on the job must be able to:

- PC23.** identify ways to optimize usage of material in various tasks/activities/processes
- PC24.** check for spills/leakages in various tasks/activities/processes
- PC25.** plug spills/leakages and escalate to appropriate authority if unable to rectify
- PC26.** check if the equipment/machine is functioning normally before commencing work and rectify wherever required
- PC27.** ensure electrical equipment and appliances are properly connected and turned off when not in use

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context
- KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same
- KU3.** evacuation procedures for workers and visitors
- KU4.** how and when to report hazards
- KU5.** potential hazards, risks and threats based on the nature of work
- KU6.** preventative and remedial actions to be taken in case of exposure to toxic material
- KU7.** various types of fire extinguisher
- KU8.** various types of safety signs and their meaning
- KU9.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.
- KU10.** relevant standards, procedures and policies related to 5S followed in the company
- KU11.** the various materials used and their storage norms
- KU12.** efficient utilisation of material and water
- KU13.** basics of electricity and prevalent energy efficient devices
- KU14.** common practices of conserving electricity
- KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU16.** usage of different colors of dustbins



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KU17. waste management techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** ask for clarifications from superior about the job requirement
- GS4.** work with supervisors/team members to carry out work related tasks
- GS5.** complete tasks efficiently and accurately within stipulated time
- GS6.** inform/report to concerned person in case of any problem
- GS7.** make timely decisions for efficient utilization of resources
- GS8.** write reports such as accident report, in at least English/regional language
- GS9.** be punctual and utilize time efficiently



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	10	14	-	6
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	2	-	1
PC2. follow safe working practices while dealing with hazards to ensure safety of self and others	2	3	-	1
PC3. use appropriate protective clothing/equipment for specific tasks and work	1	2	-	1
PC4. follow appropriate safety practices while working in and around trenches, elevated places and confined areas	2	1	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	1
PC6. carry out routine check of the machine for identifying potential hazards	1	2	-	1
PC7. report any identified breaches in health, safety and security policies and procedures to the designated person	1	2	-	1
<i>Emergencies, rescue and first aid procedures</i>	6	9	-	5
PC8. use appropriate type of fire extinguisher	1	1	-	1
PC9. apply appropriate rescue techniques during fire hazard	1	2	-	1
PC10. provide appropriate first aid procedure to victims wherever required eg.in case of bleeding, burns, choking, electric shock etc.	2	2	-	1
PC11. follow emergency procedures such as raising alarm, safe evacuation etc.	1	2	-	1
PC12. attend safety training and fire drills to respond promptly during an emergency	1	2	-	1
<i>Health and hygiene</i>	2	6	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow regular cleaning and disinfection practices at work place using appropriate techniques and materials	1	2	-	1
PC14. follow hand hygiene practices at work place using appropriate techniques and materials	1	2	-	1
PC15. report regarding the contagious illness of self or people in close contact	-	1	-	-
PC16. avoid contact with ill people and self-isolate in a similar situation	-	1	-	-
<i>Housekeeping and waste management</i>	7	12	-	5
PC17. follow the fundamentals of 5S for housekeeping	2	3	-	2
PC18. ensure good housekeeping in order to prevent hazards and accidents	1	2	-	-
PC19. store the material, tools and equipment in the correct location and in good condition	1	2	-	-
PC20. segregate waste into different categories	1	2	-	1
PC21. identify recyclable, non-recyclable and hazardous waste	1	1	-	1
PC22. dispose non-recyclable, recyclable and reusable waste appropriately at identified location	1	2	-	1
<i>Material and energy conservation</i>	5	9	-	2
PC23. identify ways to optimize usage of material in various tasks/activities/processes	1	2	-	-
PC24. check for spills/leakages in various tasks/activities/processes	1	2	-	1
PC25. plug spills/leakages and escalate to appropriate authority if unable to rectify	1	2	-	1
PC26. check if the equipment/machine is functioning normally before commencing work and rectify wherever required	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	1	-	-
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Generic
NSQF Level	3
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0009: Work effectively with others

Description

This OS unit is about communicating with colleagues/superiors and others, either in own work group or in other work groups within organisation.

Scope

The scope covers the following :

- Communicate effectively with colleagues and others
- Interact with supervisor
- Follow appropriate behaviour at work place

Elements and Performance Criteria

Communicate effectively with colleagues and others

To be competent, the user/individual on the job must be able to:

- PC1.** coordinate with colleagues to share work, as per the workload in order to achieve team goals
- PC2.** maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written
- PC3.** adjust communication styles to reflect gender and persons with disability (PwD) sensitivity
- PC4.** respect all colleagues and co-workers
- PC5.** resolve conflicts by communicating with colleagues and other departments

Interact with supervisor

To be competent, the user/individual on the job must be able to:

- PC6.** identify work requirements by receiving instructions from reporting supervisor
- PC7.** escalate problems to supervisors that cannot be handled
- PC8.** report the completed work
- PC9.** interact with the reporting supervisor about any possible hazards and safety concerns

Follow appropriate behaviour at work place

To be competent, the user/individual on the job must be able to:

- PC10.** extend help to people with Disability (PwD) at workplace, if required
- PC11.** empathize with people with disability
- PC12.** adopt a gender neutral behavior
- PC13.** adopt responsible and disciplined behaviours at the workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor



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- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with colleagues and others</i>	13	20	-	9
PC1. coordinate with colleagues to share work, as per the workload in order to achieve team goals	3	5	-	2
PC2. maintain clear communication with colleagues and others, wherever needed, through all means i.e. face-to-face, telephonic or written	5	7	-	3
PC3. adjust communication styles to reflect gender and persons with disability (PwD) sensitivity	3	4	-	2
PC4. respect all colleagues and co-workers	1	2	-	1
PC5. resolve conflicts by communicating with colleagues and other departments	1	2	-	1
<i>Interact with supervisor</i>	8	14	-	6
PC6. identify work requirements by receiving instructions from reporting supervisor	2	3	-	1
PC7. escalate problems to supervisors that cannot be handled	2	3	-	2
PC8. report the completed work	2	3	-	1
PC9. interact with the reporting supervisor about any possible hazards and safety concerns	2	5	-	2
<i>Follow appropriate behaviour at work place</i>	9	16	-	5
PC10. extend help to people with Disability (PwD) at workplace, if required	2	4	-	2
PC11. empathize with people with disability	2	4	-	1
PC12. adopt a gender neutral behavior	2	4	-	1
PC13. adopt responsible and disciplined behaviours at the workplace	3	4	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0009
NOS Name	Work effectively with others
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Generic
NSQF Level	3
Credits	1
Version	5.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0820: Understand the assigned pipeline job

Description

This unit is about understanding the requirements of the pipe line job after receiving the job details from the supervisor, seek any clarifications on the requirements of the job and identify the tools and tackles that would be needed to carry out the pipe line job.

Scope

The scope covers the following :

- Understand the assigned pipeline job in accordance with the instructions
- Understand the pipeline layout drawings
- Seek clarifications with respect to the layout of pipeline drawings, if any
- Identify the tools and tackles that are required to carry out the assigned job

Elements and Performance Criteria

Understand the assigned pipe line job in accordance with the instructions

To be competent, the user/individual on the job must be able to:

- PC1.** interpret the pipe line layout drawing and understand the pipe line job requirements ensure proper shut down of the pipe line circuit/equipment, if required
- PC2.** plan, as appropriate to carry out the pipe line job

Understand the pipe line layout drawings

To be competent, the user/individual on the job must be able to:

- PC3.** understand the layout drawing with necessary fittings
- PC4.** understand the specification of pipe line to be used for the job
- PC5.** understand the sequence of activities to complete the pipe line circuit as per lay out drawing
- PC6.** read and interpret the pipe line circuit drawings and ensure correct selection of fittings as per specification for maintaining the inclination and layout

Seek clarifications with respect to the layout of pipe line drawings, if any

To be competent, the user/individual on the job must be able to:

- PC7.** identify any clarifications that he/she wants to seek with respect to the pipe line layout/circuit drawing
- PC8.** recognize whom to contact for clarifications on the piping design
- PC9.** escalate the concern to the supervisor or shift-in-charge, if needed

Identify the tools and tackles that are required to carry out the assigned job

To be competent, the user/individual on the job must be able to:

- PC10.** identify tools, tackles & equipment required to perform the operation
- PC11.** ask helper to carry tools and pipe fittings required to the desired work site
- PC12.** report to stores / supervisor in case of non-availability of tools & tackles, pipe and pipe line fittings or stock-out



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality and damage checks to be done and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** escalation matrix for reporting identified issues
- KU4.** pipe line lay out / circuit drawings
- KU5.** tools & tackles with specified fittings to be used for the job
- KU6.** activities regarding layout and assembly of critical pipe line
- KU7.** implications of not adhering to sequence of activities and operations
- KU8.** safety precautions to be taken while working on a particular gas/steam/high pressure lines etc.
- KU9.** fabrication process (bending, assembling and threading etc.) of pipe lines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** read and interpret engineering and machine drawings
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, etc.
- GS5.** express statements, opinions or information clearly so that others can hear and understand
- GS6.** respond appropriately to queries
- GS7.** communicate with supervisor
- GS8.** communicate with team members
- GS9.** handle different machines and tools wearing protective accessories
- GS10.** position mechanical components in machines according to design requirements
- GS11.** diagnose common problems in the tools based on visual inspection, sound, temperature etc.
- GS12.** suggest improvements(if any) in maintenance processes based on experience



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand the assigned pipe line job in accordance with the instructions</i>	4	10	-	2
PC1. interpret the pipe line layout drawing and understand the pipe line job requirements ensure proper shut down of the pipe line circuit/equipment, if required	2	5	-	1
PC2. plan, as appropriate to carry out the pipe line job	2	5	-	1
<i>Understand the pipe line layout drawings</i>	8	16	-	6
PC3. understand the layout drawing with necessary fittings	2	4	-	1
PC4. understand the specification of pipe line to be used for the job	2	4	-	1
PC5. understand the sequence of activities to complete the pipe line circuit as per lay out drawing	2	4	-	2
PC6. read and interpret the pipe line circuit drawings and ensure correct selection of fittings as per specification for maintaining the inclination and layout	2	4	-	2
<i>Seek clarifications with respect to the layout of pipe line drawings, if any</i>	9	12	-	6
PC7. identify any clarifications that he/she wants to seek with respect to the pipe line layout/circuit drawing	3	4	-	2
PC8. recognize whom to contact for clarifications on the piping design	3	4	-	2
PC9. escalate the concern to the supervisor or shift-in-charge, if needed	3	4	-	2
<i>Identify the tools and tackles that are required to carry out the assigned job</i>	9	12	-	6
PC10. identify tools, tackles & equipment required to perform the operation	3	4	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ask helper to carry tools and pipe fittings required to the desired work site	3	4	-	2
PC12. report to stores / supervisor in case of non-availability of tools & tackles, pipe and pipe line fittings or stock-out	3	4	-	2
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0820
NOS Name	Understand the assigned pipeline job
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	1
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0821: Prepare for assigned pipeline job operation

Description

This unit is about inspecting the pipe line layout for scheduled maintenance or defects and testing the pipe line circuit before operation.

Scope

The scope covers the following :

- Reach the site and inspect the pipeline circuit
- Collect specified fittings and materials required for pipeline job

Elements and Performance Criteria

Reach the site and inspect the pipe line circuit

To be competent, the user/individual on the job must be able to:

PC1. reach the site with desired tools, tackles and fittings/hardware required for the job

PC2. identify the job execution methodology

Collect specified fittings and materials required for pipe line job

To be competent, the user/individual on the job must be able to:

PC3. prepare the list of spares and fittings required for completion of job

PC4. ensure that desired tools and tackles match the job specifications

PC5. ensure tools, tackles and fittings required for assembly are free from physical damage and ready to use

PC6. report damaged / defective pipe line as per the escalation matrix

PC7. prepare gap process, fabricate, repair, replacement process of pipe line

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. quality and damage checks to be done and importance of the same

KU2. risk and impact of not following defined procedures/work instructions

KU3. escalation matrix for reporting identified issues

KU4. plant layout and location of various departments

KU5. normal running hazard/ characteristics of relevant pipe line

KU6. possible causes of common problems during assembly & their remedies

KU7. tools, tackles and fittings required for the operation

KU8. standard specifications of pipe and pipe fittings

KU9. pipe line circuit isolation knowledge

KU10. hazards associated with residual substance in gas/steam/high pressure line



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** read and understand manuals, health and safety instructions, memos, reports, job cards, specifications of spare parts etc.
- GS4.** express statements, opinions or information clearly so that others can hear and understand
- GS5.** respond appropriately to queries
- GS6.** communicate with supervisor, team members, other departments e.g stores, operations, etc.
- GS7.** diagnose common problems in the tools based on visual inspection, sound, temperature etc.
- GS8.** suggest improvements(if any) in maintenance processes based on experience



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reach the site and inspect the pipe line circuit</i>	10	15	-	5
PC1. reach the site with desired tools, tackles and fittings/hardware required for the job	5	7	-	2
PC2. identify the job execution methodology	5	8	-	3
<i>Collect specified fittings and materials required for pipe line job</i>	20	35	-	15
PC3. prepare the list of spares and fittings required for completion of job	4	7	-	3
PC4. ensure that desired tools and tackles match the job specifications	4	7	-	3
PC5. ensure tools, tackles and fittings required for assembly are free from physical damage and ready to use	4	7	-	3
PC6. report damaged / defective pipe line as per the escalation matrix	4	7	-	3
PC7. prepare gap process, fabricate, repair, replacement process of pipe line	4	7	-	3
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0821
NOS Name	Prepare for assigned pipeline job operation
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ISC/N0822: Carry out the assigned pipe line operation

Description

This unit is about carrying out the operations for pipe lines in terms of rectifying the identified problem or carrying out scheduled maintenance

Scope

The scope covers the following :

- Conduct routine maintenance of pipelines and rectify /repair
- Conduct tests to ensure fitness for use
- Communicate to supervisor about completion of work

Elements and Performance Criteria

Conduct routine maintenance of pipe lines and rectify/repair

To be competent, the user/individual on the job must be able to:

- PC1.** dismantle the pipe line circuit, as needed
- PC2.** correct the defect in the pipe line circuit and replace the defective part
- PC3.** assemble the parts according to the pipe line drawings, as required
- PC4.** fasten mechanical components/ subassemblies together using pipe, pipe fittings using hand/ power tools
- PC5.** set and adjust the pipe line / fittings using pipe spanner and hand tools
- PC6.** re assemble the pipe line with fittings post correcting the defect
- PC7.** develop repair process/carry out repair process as per sop
- PC8.** handle hazardous material as per sop
- PC9.** fabrication of pipeline as per circuit drawing

Conduct tests to ensure fitness for use

To be competent, the user/individual on the job must be able to:

- PC10.** ensure the layout of pipe lines as per specified circuit drawings
- PC11.** check proper clamping to ensure the rigidity
- PC12.** pressure test of pipe line to ensure it is leak proof before handover
- PC13.** record the test results in the prescribed format of the organization

Communicate to supervisor about completion of work

To be competent, the user/individual on the job must be able to:

- PC14.** ensure all activities are complete according to checklist
- PC15.** communicate to supervisor on completion of given job in case of any deviations from checklist

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** quality and damage checks to be done and importance of the same
- KU2.** contact person across departments for spare parts, information etc.
- KU3.** escalation matrix for reporting identified issues
- KU4.** risk and impact of not following defined procedures/work instructions
- KU5.** assembling techniques such as aligning, bending, fixing, mechanical jointing, threaded jointing, sealing etc.
- KU6.** steps required to assemble/ dismantle pipe line circuit as per pipe line lay out
- KU7.** checks that need to be made to ensure that the pipe line circuit is safe and ready to use
- KU8.** proper layout and rigid fixing for trouble free operation
- KU9.** possible causes of common problems during assembly & their remedies
- KU10.** units of measurement
- KU11.** response to emergencies e.g. water supply failures, fire and system failures
- KU12.** use of measuring instruments e.g. steel scale, steel tape, micrometer, vernier etc.
- KU13.** compilation of pressure test results in prescribed format

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** read and interpret engineering and machine drawings
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, etc.
- GS5.** express statements, opinions or information clearly so that others can hear and understand
- GS6.** respond appropriately to queries
- GS7.** communicate with supervisor, team members, other departments e.g stores, operations, etc.
- GS8.** diagnose common problems in the tools based on visual inspection, sound, temperature etc.
- GS9.** suggest improvements(if any) in maintenance processes based on experience



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct routine maintenance of pipe lines and rectify/repair</i>	18	30	-	10
PC1. dismantle the pipe line circuit, as needed	2	3	-	1
PC2. correct the defect in the pipe line circuit and replace the defective part	2	4	-	2
PC3. assemble the parts according to the pipe line drawings, as required	2	3	-	1
PC4. fasten mechanical components/ subassemblies together using pipe, pipe fittings using hand/ power tools	2	4	-	1
PC5. set and adjust the pipe line / fittings using pipe spanner and hand tools	2	3	-	1
PC6. re assemble the pipe line with fittings post correcting the defect	2	4	-	1
PC7. develop repair process/carry out repair process as per sop	2	3	-	1
PC8. handle hazardous material as per sop	2	3	-	1
PC9. fabrication of pipeline as per circuit drawing	2	3	-	1
<i>Conduct tests to ensure fitness for use</i>	8	12	-	6
PC10. ensure the layout of pipe lines as per specified circuit drawings	2	3	-	2
PC11. check proper clamping to ensure the rigidity	2	3	-	1
PC12. pressure test of pipe line to ensure it is leak proof before handover	2	3	-	2
PC13. record the test results in the prescribed format of the organization	2	3	-	1
<i>Communicate to supervisor about completion of work</i>	4	8	-	4



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. ensure all activities are complete according to checklist	2	4	-	2
PC15. communicate to supervisor on completion of given job in case of any deviations from checklist	2	4	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0822
NOS Name	Carry out the assigned pipe line operation
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	2
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0823: Activities specific to pipeline operation

Description

This unit is about carrying out the operations that are specific to a pipe line fitter.

Scope

The scope covers the following :

- Ensure the correctly colour coded pipes are used
- Fabrication of Pipeline

Elements and Performance Criteria

Ensure the correctly colour coded pipes are used

To be competent, the user/individual on the job must be able to:

- PC1.** understand the colour coding system of pipes
- PC2.** identify which colour pipe is used with which media (water line, gas line, steam line, etc.)
- PC3.** correctly use the pipes in the given media
- PC4.** rectify any defects in the pipe line system
- PC5.** test the pipeline and ensure fitness
- PC6.** report to the supervisor in case of any problem that has to be escalated

Fabrication of PipeLine

To be competent, the user/individual on the job must be able to:

- PC7.** understand the process energy isolation for the pipe line
- PC8.** understand layout of the pipe line circuit
- PC9.** handle hazardous waste of the pipe line

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality and damage checks to be done and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** escalation matrix for reporting identified issues
- KU4.** piping colour codes along with relevant media and connected safety aspects
- KU5.** farvel system for automatic grease line
- KU6.** implication of working on high pressure water line/gas line/steam line/oxygen line etc.
- KU7.** pressure testing to ensure the pipe line if leak proof
- KU8.** checks that need to be made to ensure that equipment is safe and ready to use
- KU9.** limits, fits and tolerances
- KU10.** nature of fluid / gas being carried through pipeline



Qualification Pack

KU11. isolating units through valves, blanks, liquid seals etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences and express ideas clearly through written communication
- GS2.** fill up appropriate technical forms, process charts, activity logs in required format of the company
- GS3.** read and interpret engineering and machine drawings
- GS4.** read and understand manuals, health and safety instructions, memos, reports, job cards, etc.
- GS5.** express statements, opinions or information clearly so that others can hear and understand
- GS6.** respond appropriately to queries
- GS7.** communicate with supervisor
- GS8.** communicate with team members
- GS9.** diagnose common problems in the tools based on visual inspection, sound, temperature etc.
- GS10.** suggest improvements(if any) in maintenance processes based on experience



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure the correctly colour coded pipes are used</i>	20	32	-	14
PC1. understand the colour coding system of pipes	4	6	-	3
PC2. identify which colour pipe is used with which media (water line, gas line, steam line, etc.)	4	5	-	2
PC3. correctly use the pipes in the given media	3	6	-	3
PC4. rectify any defects in the pipe line system	3	5	-	2
PC5. test the pipeline and ensure fitness	3	5	-	2
PC6. report to the supervisor in case of any problem that has to be escalated	3	5	-	2
<i>Fabrication of PipeLine</i>	10	18	-	6
PC7. understand the process energy isolation for the pipe line	3	6	-	2
PC8. understand layout of the pipe line circuit	4	6	-	2
PC9. handle hazardous waste of the pipe line	3	6	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0823
NOS Name	Activities specific to pipeline operation
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Mechanical Maintenance
NSQF Level	3
Credits	3
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	30	50	-	20	100	10
ISC/N0009.Work effectively with others	30	50	-	20	100	10
ISC/N0820.Understand the assigned pipeline job	30	50	-	20	100	10
ISC/N0821.Prepare for assigned pipeline job operation	30	50	-	20	100	10
ISC/N0822.Carry out the assigned pipe line operation	30	50	-	20	100	35
ISC/N0823.Activities specific to pipeline operation	30	50	-	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	200	330	-	120	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.